Novelty Check and Literature Review

Sparse-View 3D Gaussian Splatting with AI Foundation Models | Search date: 3 November 2026
(cross-checked with Elicit, Consensus and Perplexity)
Farida Gaber and Mona Yasser | Supervisor: Dr. Wedad Sallam

1. Purpose

Before starting the experiments, we checked whether our research idea, or a very similar one, has already been published. This document lists the closest papers, what they did, what they did not do, and our honest conclusion about the research gap.

2. The question we checked

Has anyone already separated the effect of (1) AI-predicted camera poses, (2) AI-predicted 3D points, and (3) joint pose refinement in sparse-view 3D Gaussian Splatting, across different numbers of input views, on the Mip-NeRF 360 dataset?

3. Short answer

No paper was found that does this full study. Several papers do small parts of it. The most related papers are **VGGT-X** [7] (dense views only) and **SPARS3R** [4]. Three research tools (Elicit, Consensus, Perplexity) gave the same verdict: *partly done*. Our study is therefore an **extension of existing work with a new controlled experimental setup** (an analysis / ablation study), not a completely new idea and not a new algorithm.

4. Key terms (simple definitions)

Term	Simple meaning
3D Gaussian Splatting (3DGS)	A method that turns many photos of a scene into a 3D model made of small coloured 'blobs' (Gaussians), which can be rendered from new viewpoints.
Camera pose	Where the camera was and which direction it was pointing for each photo.
COLMAP / SfM	Classic software (Structure-from-Motion) that computes camera poses by matching points between photos. Works well with many photos, often fails with very few.
Sparse-view	Training with only a few photos (for example 3 to 24) instead of 100+.
Foundation model (DUSt3R, MASt3R, VGGT, pi^3)	Large pretrained neural networks that predict camera poses and 3D points directly from images.
Joint pose refinement	Correcting the camera poses during 3DGS training, together with the 3D model (used by InstantSplat).
Reference poses	Camera poses computed by COLMAP from ALL photos of the scene (provided with the dataset). Very accurate but not perfect, and not available in a real few-photo situation.
PSNR / SSIM / LPIPS	Standard scores of how similar a rendered image is to the real photo. Higher PSNR/SSIM is better; lower LPIPS is better.
ATE / RPE	Camera pose error: how far the estimated cameras are from the reference cameras.

5. Closest papers (ranked by relevance)

#	Paper	What it does	Data / views	What it does NOT do (gap)
1	VGGT-X [7] (2025)	Replaces COLMAP with VGGT for 3DGS. Its ablation (Table 4) tests pose refinement on/off and random vs AI-predicted points.	Mip-NeRF 360 (ablation), Tanks and Temples, CO3D; dense views (up to 311 photos)	Dense views only. No reference-pose swap with the same points. Does not test 3-9 photos.
2	SPARS3R [4] (CVPR 2025)	Aligns MASt3R dense points with COLMAP sparse points to initialize 3DGS. Compares COLMAP points vs MASt3R points; measures pose error of InstantSplat.	Mip-NeRF 360, 9 scenes, one view count	No reference-pose condition, no pose-refinement on/off test, only one view count.
3	SDI-GS [19] (2025)	Same pipeline as InstantSplat (MASt3R poses + points, joint pose refinement) with segmentation-based point selection.	Mip-NeRF 360 (9 scenes), Tanks and Temples, MVImgNet; 3 / 6 / 12 views	No reference-pose, random-point or refinement on/off test (only ablates label dimension).
4	InstantSplat [2] (v6, 2025)	MASt3R predicts poses + dense points, then poses and Gaussians are optimized together. Compares with COLMAP+3DGS and pose-free methods.	Tanks and Temples, MVImgNet; 3 / 6 / 12 views	Ablation adds parts one by one, on Tanks and Temples, one view count. No reference-pose or random-point condition.
5	Free360 [3] (CVPR 2025)	Layered Gaussians + video diffusion, from unposed images.	Mip-NeRF 360, 8 scenes, 3 / 6 / 9 views	Proposes a new method; uses InstantSplat only as a baseline. No separation of poses / points / refinement.
6	MS-GS [5] (NeurIPS 2025)	Sparse-view 3DGS for photos with different lighting. Points out that most papers use poses computed from all dense images.	Mip-NeRF 360 (8 scenes), 20 views	One view count only; focus on appearance changes.
7	Foroutan et al. [13] (2024)	Random vs SfM vs NeRF-derived points with fixed COLMAP cameras. Shows random results depend strongly on how the points are sampled.	Mip-NeRF 360, OMMO; dense	Dense views; poses always known; no foundation-model points.
8	Initialize to Generalize [20] (2025)	Controls initialization strength (SfM from 6 / 12 / all views) separately from training regularizers; finds initialization is the decisive factor.	Mip-NeRF 360 and LLFF; sparse; dataset poses	Points only; no AI poses, no refinement test.
9	Pi-GS [9] (Feb 2026)	Uses the pi^3 foundation model to initialize sparse-view 3DGS.	Mip-NeRF 360, 3 and 12 views; InstantSplat is a baseline	Ablation is cumulative on one Tanks and Temples scene. No separation of poses / points / refinement.
10	Role of Initialization [21] (ACCV 2026)	Compares SfM, laser, monocular depth, EDGS and Depth Anything 3 initializations with known cameras.	Mip-NeRF 360, Tanks and Temples, ScanNet++, ETH3D; dense	Dense views; known cameras; no pose or refinement test.
11	He et al. [6] (2025)	Dense stereo initialization + diffusion pseudo-views + depth regularization.	Tanks and Temples (2/3/6); Mip-NeRF 360 (4/6/9) in supplement	New method paper; no separating ablation.
12	3R-GS [8] (2025)	MASt3R-SfM poses + pose refinement during 3DGS training.	Mip-NeRF 360 (4 scenes), dense	Dense views only; no full refinement-off test.
13	Dalal et al. [10] (Jul 2026)	VGGT-X initialization + pose refiner + depth loss.	Mip-NeRF 360 (4 scenes), 50-60 views	Not sparse; ablation only removes MCMC / depth.

#	Paper	What it does	Data / views	What it does NOT do (gap)
14	FSGS [12] (2024)	Sparse-view 3DGS with depth priors.	Mip-NeRF 360, 24 views	Uses poses computed from all images (the issue raised by MS-GS).
15	CF-3DGS [11] (2024)	COLMAP-free 3DGS from ordered video frames.	Tanks and Temples, CO3D (dense video)	Needs ordered video; not suitable for sparse unordered views.
16	NoPoSplat [14] / SPFSplat	Feed-forward networks trained to output Gaussians from unposed images; ablate backbone initialization.	RealEstate10K, ACID and similar	Different setting (trained networks, other datasets).

6. Useful numbers from the literature

These numbers were read from the original papers' tables. They give us a reproduction target and a starting hypothesis.

6.1 InstantSplat on Mip-NeRF 360 (from Free360 [3], Table 1) - our reproduction target

Training views	PSNR (higher better)	SSIM (higher better)	LPIPS (lower better)
3	14.10	0.296	0.529
6	15.83	0.351	0.480
9	17.16	0.402	0.443

6.2 SPARS3R [4] on Mip-NeRF 360 with 12 views

Setting	PSNR	SSIM	LPIPS
3DGS with COLMAP (SfM) points	16.6	0.388	0.458
3DGS with MASt3R points	15.9	0.293	0.463
SPARS3R (aligned MASt3R + SfM)	18.9	0.500	0.327

SPARS3R also reports relative pose error on Mip-NeRF 360 (translation / rotation): InstantSplat 2.049 / 2.555 versus COLMAP + MASt3R matching 0.252 / 0.412, and notes that test-pose optimization "can get stuck if the initial displacement is too large, as is sometimes the case in InstantSplat."

6.3 InstantSplat on Tanks and Temples (from InstantSplat [2], Table 2)

Method	Time (3 views)	SSIM 3 / 6 / 12 views	LPIPS 3 / 6 / 12 views
COLMAP + 3DGS	4 min 28 s	0.376 / 0.592 / 0.716	0.513 / 0.343 / 0.251
InstantSplat (XL)	25.4 s	0.762 / 0.845 / 0.879	0.163 / 0.117 / 0.107

6.4 Dense views (VGGT-X [7], Mip-NeRF 360): COLMAP-initialized 3DGS reaches PSNR 27.39; the COLMAP-free VGGT-X pipeline reaches 26.40. Even with many images, a gap remains. VGGT-X's ablation (Table 4, dense views) shows that turning pose refinement on raises PSNR from 22.37 to 26.28, and that random points (26.28) work about as well as VGGT-derived points (25.88 / 26.40). Our study tests whether this still holds with only 3-9 photos.

7. What is already done and what is still open

Part of our study	Done before?	By whom / notes
COLMAP-3DGS vs COLMAP-free comparison	Yes	InstantSplat [2], He et al. [6], VGGT-X [7]
Several view counts on Mip-NeRF 360	Yes	Free360 [3] (3/6/9), SDI-GS [19], He et al. [6], Pi-GS [9]
COLMAP points vs MASt3R points (same poses)	Partly	SPARS3R [4], one view count
Pose refinement on vs off	Partly	VGGT-X [7], dense views only; InstantSplat [2], one view count
AI points vs random points	Partly	VGGT-X [7] and Foroutan et al. [13], dense views only
Reference poses vs AI poses in the same pipeline	Not found	MS-GS [5] raises the issue at 20 views, different focus
All three together, at 3 / 6 / 9 views, Mip-NeRF 360	Not found	Our planned contribution

8. Conclusion

- Our original idea (compare COLMAP-3DGS with a COLMAP-free method at several view counts on Mip-NeRF 360) has **already been done** in large part. On its own it would be a replication.
- Two of our three switches (pose refinement on/off, AI vs random points) were tested by VGGT-X [7], but **only with dense views**. The reference-vs-AI pose swap in the same pipeline, and all three switches together with 3 / 6 / 9 views, were **not found**.
- Our study is an **extension** of VGGT-X [7], SPARS3R [4] and InstantSplat [2]. Its main question becomes: **do the dense-view findings still hold with only 3-9 photos?**
- Hypotheses to test: with few photos, AI points may matter much more than with dense views, and pose refinement may fail when the starting pose error is large [4].
- Classification: **existing method + new controlled experimental setup** (analysis / ablation study).

9. Cross-check with research tools

The same question was given to three AI research tools. Every paper they named was opened and checked against the original text.

Tool	Verdict	Notes
Elicit	Partly done	Closest: VGGT-X. One wrong claim found (a RAIN-GS table that does not exist).
Consensus	Partly done	Nothing closer than the papers above. Gave useful method warnings (reference poses, random-point definition, training images only).
Perplexity	Partly done	Found SDI-GS [19]. Two wrong details found (InstantSplat datasets; one paper title).

10. Limits of this check (honest notes)

- About 30 papers were checked in detail (methods and experiments), including every paper suggested by Elicit, Consensus and Perplexity. The field is very fast, so a paper may exist that we did not find.
- DPR-Splat [22] (IROS 2025; sparse-view, MASt3R pose refinement) could not be read yet because it is not freely available. It must be checked before submission.
- Some details were read from HTML versions of the papers. All numbers will be re-checked in the original PDFs before they are cited in the final paper.
- Not finding an identical paper does not prove novelty. The supervisor's opinion and a final search before submission are still needed.

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